End-to-End Retail Data Analytics Project

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Project Objective & Scope

This project analyzes a retail transaction dataset covering the period December 2010 to December 2011. The dataset contains invoice-level purchase records including InvoiceNo, InvoiceDate, CustomerID, Quantity, UnitPrice, and Country. A Revenue variable was calculated as Quantity multiplied by UnitPrice to measure order performance.

The objective of this project was to perform an end-to-end data analytics workflow, starting from data cleaning and preparation, moving through exploratory analysis and dashboard development, and finally validating business insights using statistical hypothesis testing.

The analysis aimed to identify seasonal revenue patterns, evaluate product performance, understand customer retention behavior through cohort analysis, and determine whether seasonal revenue growth was driven by higher order value or increased transaction volume.

The final outcome of this project was to translate raw transactional data into validated, business-focused insights supported by visualization, statistical testing, and actionable strategic recommendations.



Data Preparation & Processing (Task 1)

The first phase of the project focused on preparing the raw retail dataset for analysis. The dataset initially contained missing values, duplicate records, and inconsistent formats that required cleaning before meaningful insights could be extracted.

During this stage, missing values in key columns such as CustomerID were identified and handled appropriately. Duplicate invoice records were removed to ensure data accuracy. Data types were standardized, particularly for InvoiceDate, to enable time-based analysis.

A new variable called Revenue was created by multiplying Quantity and UnitPrice. This transformation enabled order-level revenue analysis and became a core metric throughout the project.

The cleaned dataset provided a structured and reliable foundation for exploratory analysis, dashboard development, and statistical testing in the later stages of the internship.

Exploratory Data Analysis (Task 2)



After preparing the cleaned dataset, the next phase focused on exploratory data analysis to identify patterns, trends, and key performance drivers.
Monthly revenue analysis revealed clear seasonal patterns, with significant revenue peaks observed during September, October, and November. These findings suggested strong seasonal demand toward the end of the year.
Product-level analysis showed that a small group of products contributed a large portion of total revenue, indicating revenue concentration among high-performing items.
Cohort analysis was performed to evaluate customer retention over time. The results showed that most customers were one-time buyers, while a smaller segment of repeat customers contributed recurring revenue.
Overall, exploratory analysis provided a structured understanding of revenue behavior, customer trends, and product performance, forming the basis for dashboard development and statistical validation in later stages.

Interactive Dashboard Development (Task 3)

In the third phase of the project, the analytical insights were translated into an interactive business dashboard using Power BI.
Key performance indicators were defined, including Total Revenue, Total Customers, Total Orders, and Average Order Value. These KPIs provided a high-level summary of overall business performance.
A monthly revenue trend visualization was developed to monitor seasonality and identify peak demand periods. A top-products analysis was included to highlight revenue concentration among high-performing items.
Cohort retention results were integrated into the dashboard to provide visibility into customer lifecycle behavior and repeat purchasing patterns.
The dashboard enabled dynamic exploration of revenue, customer activity, and product performance, transforming analytical findings into a decision-support tool suitable for business stakeholders.

Statistical Validation (Task 4)

In the final analytical phase of the project, statistical hypothesis testing was conducted to validate the seasonal revenue patterns identified during exploratory analysis.

The key business question was whether the higher total revenue observed during peak months was driven by an increase in average order value.

To test this, an Independent Two-Sample T-Test was performed comparing average order revenue between high-revenue months (September, October, November) and low-revenue months (January, February).

The test produced a p-value of 0.2238, which is greater than the significance level of 0.05. Therefore, the null hypothesis was not rejected.

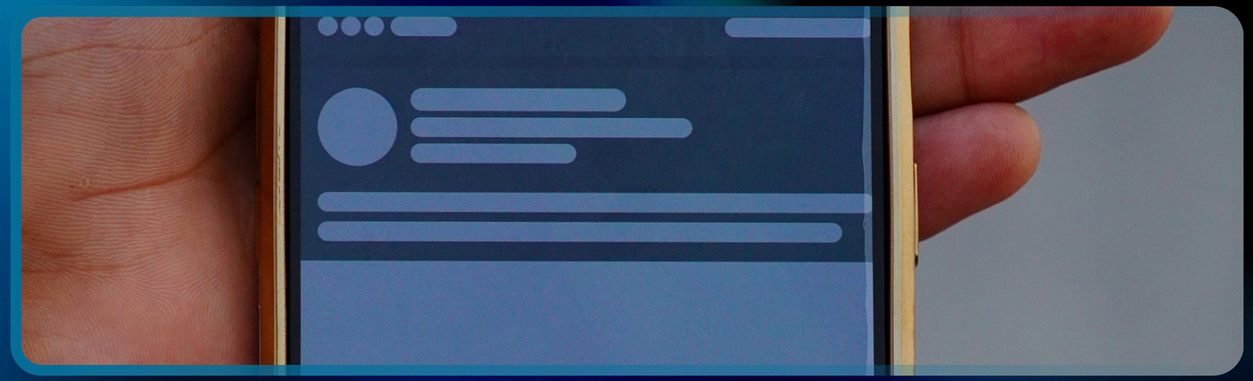
This result indicates that there is no statistically significant difference in average order value between seasonal periods. The increase in total revenue during peak months is likely driven by higher transaction volume rather than increased spending per order.

Key Insights & Business Impact

The end-to-end analysis revealed several important business insights.
Revenue shows strong seasonal peaks during the final quarter of the year, particularly in September through November, indicating increased customer activity during this period.
Product-level analysis demonstrated that a small group of high-performing items contributes a significant portion of total revenue, highlighting opportunities for targeted promotions and inventory prioritization.
Cohort analysis revealed that the majority of customers are one-time buyers, while a smaller segment of repeat customers drives recurring revenue, emphasizing the importance of retention strategies.

Statistical validation confirmed that seasonal revenue growth is driven by increased transaction volume rather than higher order value.
Overall, this project demonstrates how structured data cleaning, exploratory analysis, dashboard development, and statistical testing can transform raw transactional data into actionable, business-focused insights that support strategic decision-making.

Strategic Recommendations



- Targeted promotional campaigns should be implemented in the months leading up to September, October, and November to capitalize on seasonal demand patterns.
  Given that a small group of products contributes significantly to total revenue, these high-performing items should be prioritized in advertising strategies, bundling offers, and inventory planning.

- Cohort analysis indicates that customer retention is relatively low, with most customers making only a single purchase. Therefore, implementing loyalty programs, personalized offers, and follow-up engagement strategies can help increase repeat purchases.

- Since revenue growth during peak months is driven primarily by increased transaction volume, the business should focus on maximizing customer acquisition and marketing efforts before and during high-demand periods.

Tools & Technologies Used



- Python was used as the primary programming language for data processing and analysis.
- Pandas and NumPy were used for data cleaning, transformation, and aggregation tasks.
- Matplotlib and Seaborn were applied to create visualizations for exploratory data analysis.
- SciPy was used to perform statistical hypothesis testing, including the Independent Two-Sample T-Test.
- Power BI was used to design and develop an interactive business dashboard that visualized key performance indicators and revenue trends.

Final Conclusion



- The analysis identified clear seasonal revenue patterns, highlighted product-level performance concentration, and revealed opportunities for improving customer retention.
- Statistical testing confirmed that seasonal revenue growth is driven primarily by increased transaction volume rather than higher average order value.
- By combining data visualization, statistical reasoning, and business interpretation, this project transforms raw transactional data into actionable insights that support strategic decision-making.
- Overall, this internship project reflects the practical application of analytical skills, statistical validation, and business intelligence tools to solve real-world data problems.

Thank **You.**